

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I R. PRANATHI(192525076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

ANS:

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet on a pay-as-you-use basis.

Characteristics:

- On-demand self-service – Users can access resources without human intervention.
- Broad network access – Services are available over the internet using computers, tabs, or mobile phones.
- Resource pooling – Multiple users share the same infrastructure securely.
- Rapid elasticity – Resources can be increased or decreased quickly based on demand.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

ANS:

- **Hardware Evolution:** Mainframes → Personal Computers → Distributed Systems → Data Centers.
- **Internet Evolution:** High-speed internet enabled global connectivity.
- **Virtualization:** Allowed multiple virtual machines on one physical server, improving resource utilization.
- **Web Services:** SOAP and REST enabled applications to communicate over the internet.
- **Modern Cloud:** These technologies led to scalable cloud platforms such as AWS, Azure, and Google Cloud.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

ANS:

ROLE:

- Creates multiple virtual machines on one physical server.
- Improves hardware utilization.
- Reduces infrastructure cost.
- Enables easy resource management and scalability.

DIFFERENCE BETWEEN VIRTUALIZATION AS AN ENABLING TECHNOLOGY AND CLOUD COMPUTING AS A SERVICE MODEL:

Virtualization	Cloud Computing
• Technology that creates virtual machines • Focuses on resource abstraction • Foundation of cloud	• Service that delivers computing resources over the internet • Focuses on service delivery • Uses virtualization to provide services

4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.

ANS:

COMPARISON BETWEEN SOAP-BASED AND REST-BASED WEB SERVICES:

SOAP	REST
• Protocol • Uses XML only • More secure but complex • Suitable for enterprise applications	• Architectural style • Supports JSON, XML, and other formats • Lightweight and faster • Suitable for cloud and web/mobile applications

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

ANS:

DIFFERENCE BETWEEN IaaS, SaaS, XaaS:

Service Model	Description	Example
• IaaS	Provides virtual servers, storage, and networking	Amazon EC2
• PaaS	Provides a platform to develop and deploy applications	Google App Engine
• SaaS	Provides ready-to-use software over the internet	Microsoft 365
• XaaS	Any service delivered over the internet	Dropbox (SaaS)